Calculus and Linear Algebra

Second Derivative, Concavity and Inflection Points

Lecture 22

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Lecture outline

- The second derivative as the rate of change of the slope.
- Concavity and convexity from the sign of f''.
- Inflection points and changes of concavity.
- Second derivative test for local extrema.
- First and second derivatives together.
- Applications and exercises.

From f' to f''

The first derivative measures the slope of f:

$$f'(x) = \text{instantaneous rate of change of } f.$$

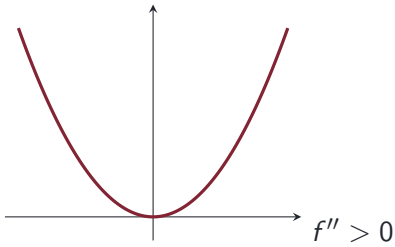
The second derivative measures how that slope changes:

$$\boxed{f''(x) = \frac{d}{dx} f'(x).}$$

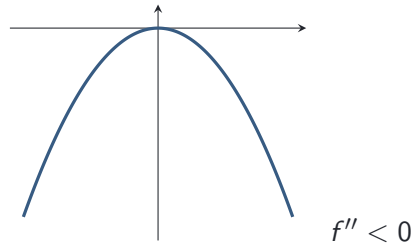
If $f'' > 0$, slopes are increasing; if $f'' < 0$, slopes are decreasing.

A geometric comparison

Convex / concave up



Concave / concave down



Terminology

On an interval I:

- f is **convex** (or concave up) if its graph bends upward.
- f is **concave** (or concave down) if its graph bends downward.

For twice differentiable functions,

$$\boxed{f''(x) > 0 \Rightarrow f \text{ is convex}},$$

$$\boxed{f''(x) < 0 \Rightarrow f \text{ is concave}}.$$

Interpretation

Convexity is about the behaviour of the slope, not simply whether the function rises or falls.

Increasing but concave down? Yes

Consider

$$f(x) = \sqrt{x}, \qquad x > 0.$$

Then

$$f'(x) = \frac{1}{2\sqrt{x}} > 0,$$

so f is increasing, but

$$f''(x) = -\frac{1}{4x^{3/2}} < 0,$$

so it is concave down.

Two independent questions

f' tells whether the graph rises or falls; f'' tells how its slope changes.

Decreasing but convex? Also yes

For

$$f(x) = e^{-x},$$

we have

$$f'(x) = -e^{-x} < 0, \qquad f''(x) = e^{-x} > 0.$$

Therefore the function is decreasing but convex on all of $\mathbb{R}$. The slope is negative, but becomes less negative as x increases.

Inflection point: definition

Definition A point $(x_0, f(x_0))$ is an **inflection point** if the concavity of the graph changes at x_0.

Essential condition

The concavity must actually change from one side to the other.

Often $f''(x_0) = 0$, but this alone is not sufficient.

Prototype: $f(x) = x^3$

$$f'(x) = 3x^2, \qquad f''(x) = 6x.$$

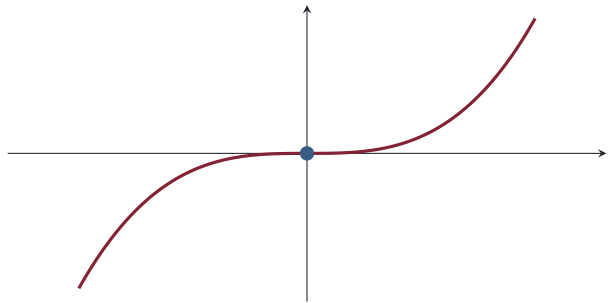
Thus

$$f''(x) < 0 \text{ for } x < 0, \qquad f''(x) > 0 \text{ for } x > 0.$$

The concavity changes at 0, so

$$\boxed{(0,0) \text{ is an inflection point.}}$$

Graph of an inflection point



The graph is concave down on the left and convex on the right.

$f''(x_0) = 0$ does not guarantee inflection

Take

$$f(x) = x^4.$$

Then

$$f''(x) = 12x^2, \qquad f''(0) = 0.$$

But

$$f''(x) > 0 \quad \text{for } x \neq 0.$$

There is no change of concavity, hence

$$\boxed{(0,0) \text{ is not an inflection point.}}$$

How to find concavity and inflection points

1. Compute f''.
2. Find points where $f'' = 0$ or f'' does not exist.
3. Respect the domain of f.
4. Use these points to divide the domain into intervals.
5. Determine the sign of f'' on each interval.
6. Record every actual change of concavity.

Example 1: polynomial

Exercise Study concavity and inflection points of

$$f(x) = x^3 - 3x^2 + 2.$$

$$f'(x) = 3x^2 - 6x, \qquad f''(x) = 6x - 6 = 6(x - 1).$$

The candidate point is

$$x = 1.$$

Example 1: sign of f''

$$f''(x) < 0 \quad \text{for } x < 1, \qquad f''(x) > 0 \quad \text{for } x > 1.$$

Hence

$$\text{concave on } (-\infty, 1), \qquad \text{convex on } (1, +\infty).$$

Since

$$f(1) = 0,$$

we obtain the inflection point

$$\boxed{(1, 0).}$$

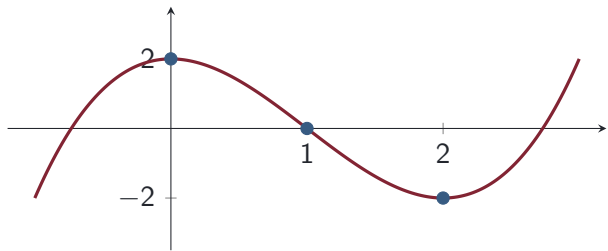
First and second derivative together

For

$$f(x) = x^3 - 3x^2 + 2,$$

we already know:

- local maximum at $(0, 2)$;
- local minimum at $(2, -2)$;
- inflection point at $(1, 0)$.



Second derivative test for extrema

Suppose

$$f'(x_0) = 0.$$

If

$$f''(x_0) > 0,$$

then f has a local minimum at x_0. If

$$f''(x_0) < 0,$$

then f has a local maximum at x_0.

If $f''(x_0) = 0$

The test is inconclusive. Return to the first derivative sign test or another argument.

Why the second derivative test makes sense

At a stationary point x_0:

- $f''(x_0) > 0$ means the slope f' is increasing through zero: negative $\to$ positive, giving a minimum.
- $f''(x_0) < 0$ means the slope f' is decreasing through zero: positive $\to$ negative, giving a maximum.

$f'' > 0$: f' $\nearrow$ through 0

$f'' < 0$: f' $\searrow$ through 0

Example 2: second derivative test

Let

$$f(x) = x^4 - 4x^2.$$

$$f'(x) = 4x(x^2 - 2),$$

so the stationary points are $0, \pm\sqrt{2}$. Also

$$f''(x) = 12x^2 - 8.$$

Then

$$f''(0) = -8 < 0$$

so $x = 0$ is a local maximum, while

$$f''(\pm\sqrt{2}) = 16 > 0$$

so $x = \pm\sqrt{2}$ are local minima.

When the test fails

For

$$f(x) = x^4,$$

we have

$$f'(0) = 0, \qquad f''(0) = 0.$$

The second derivative test gives no conclusion. Yet $x = 0$ is an absolute minimum because

$$x^4 \geq 0.$$

Lesson

An inconclusive test does not mean there is no extremum.

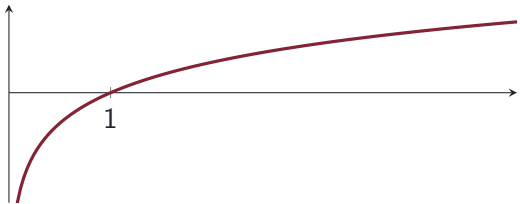
Example 3: logarithm

Consider

$$f(x) = \ln x, \qquad x > 0.$$

$$f'(x) = \frac{1}{x} > 0, \qquad f''(x) = -\frac{1}{x^2} < 0.$$

Therefore $\ln x$ is increasing and concave on $(0, +\infty)$.



Example 4: exponential

For

$$f(x) = e^x,$$

$$f'(x) = e^x > 0, \qquad f''(x) = e^x > 0.$$

Hence e^x is increasing and convex on $\mathbb{R}$.

Useful contrast

$\ln x$ is increasing and concave; e^x is increasing and convex.

Economic interpretation of convexity

Suppose $C(q)$ is a cost function.

$$C'(q) = \text{marginal cost.}$$

If

$$C''(q) > 0,$$

then marginal cost is increasing with output. For a revenue or utility function, concavity often represents diminishing marginal gains:

$$f''(x) < 0 \Rightarrow f'(x) \text{ decreases as } x \text{ increases.}$$

Exercise: concavity

Exercise Let

$$f(x) = x^4 - 4x^3.$$

Find:

1. intervals of convexity and concavity;
2. all inflection points.

Exercise: solution

Solution

$$f'(x) = 4x^3 - 12x^2, \qquad f''(x) = 12x^2 - 24x = 12x(x-2).$$

Candidates: 0 and 2.

$$f'' > 0 \text{ on } (-\infty, 0) \cup (2, +\infty),$$

$$f'' < 0 \text{ on } (0, 2).$$

Thus the inflection points are

$$(0, 0), \qquad (2, -16).$$

Exercise: combine f' and f''

Exercise For

$$f(x) = x^3 - 6x^2 + 9x,$$

determine:

- increasing/decreasing intervals;
- local extrema;
- convex/concave intervals;
- inflection points.

This is already the core of a complete function study.

Exercise: solution, first derivative

$$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3).$$

Hence

$$f' > 0 \text{ on } (-\infty, 1) \cup (3, +\infty),$$

$$f' < 0 \text{ on } (1, 3).$$

Therefore

$(1, 4)$ is a local maximum, $(3, 0)$ is a local minimum.

Exercise: solution, second derivative

$$f''(x) = 6x - 12 = 6(x - 2).$$

Therefore

$$f'' < 0 \text{ for } x < 2, \qquad f'' > 0 \text{ for } x > 2.$$

So the function is concave on $(-\infty, 2)$ and convex on $(2, +\infty)$, with inflection point

$$\boxed{(2, 2).}$$

A complete qualitative table

For $f(x) = x^3 - 6x^2 + 9x$:

x	$(-\infty, 1)$	1	$(1, 2)$	2	$(2, 3)$	3	$(3, \infty)$
f'	$+$	0	$-$	$-$	$-$	0	$+$
f''	$-$	$-$	$-$	0	$+$	$+$	$+$

This table contains enough information to reconstruct the main shape of the graph.

Common mistakes

- Using $f'' = 0$ as the definition of an inflection point.
- Confusing increasing/decreasing with convex/concave.
- Applying the second derivative test when $f'(x_0) \neq 0$.
- Declaring "no extremum" when $f''(x_0) = 0$.
- Forgetting domain restrictions before constructing sign charts.

Independent practice

Exercise Analyze first and second derivatives for:

$$\text{A: } f(x) = \frac{x^2}{x^2+1},$$

$$\text{B: } g(x) = xe^{-x},$$

$$\text{C: } h(x) = \ln(x^2+1).$$

Identify monotonicity, extrema, concavity and any inflection points.

What to remember

$$\boxed{f' > 0 / f' < 0 \quad \Rightarrow \quad \text{monotonicity}}$$

$$\boxed{f'' > 0 / f'' < 0 \quad \Rightarrow \quad \text{convexity/concavity}}$$

$$\boxed{\text{inflection point} = \text{actual change of concavity}}$$

Next lecture

We combine every tool developed so far into a complete function study.